

```
exercise3.py x
1 class Characters:
2     def __init__(self, phrases):
3         self.phrases = phrases
4     def len(self):
5         return sum([len(phrase) for phrase in self.phrases])
6     def __lt__(self, other):
7         return self.len() < other.len()
8     def __gt__(self, other):
9         return self.len() > other.len()
10    def __eq__(self, other):
11        return self.len() == other.len()
12
13    sample_phrases1 = ['cat in the hat', 'green eggs and ham', 'the lorax']
14    sample_phrases2 = ['the taming of the shrew', 'hamlet', 'othello']
15
16    c1 = Characters(sample_phrases1)
17    c2 = Characters(sample_phrases2)
18    print(c1 > c2) # prints 'True'
19    print(c1 < c2) # prints 'False'
20    print(c1 == c1) # prints 'True'
```

Polymorphism Exercise 3

Exercise 3

Create the class `Characters` which has the attribute `phrases` which is a list of strings passed as a parameter. Overload the `<`, `>`, and `==` operators so that you can make comparisons based on the total number of characters in the string.

Expected Output

If you were to run the following code with your class, the output would be `True`, `False`, and `True`.

```
sample_phrases1 = ['cat in the hat', 'green eggs and ham', 'the lorax']
sample_phrases2 = ['the taming of the shrew', 'hamlet', 'othello']

c1 = Characters(sample_phrases1)
c2 = Characters(sample_phrases2)
print(c1 > c2) # prints 'True'
print(c1 < c2) # prints 'False'
print(c1 == c1) # prints 'True'
```

TRY IT

True
False
True

Submit your code for evaluation

To find the total number of characters, iterate through each string in the list and increment an accumulator variable with the length of each string. This functionality is handled by a separate method; this is not required. Here is one possible solution:

```
class Characters:
    def __init__(self, phrases):
        self.phrases = phrases

    def total_characters(self):
        total = 0
        for phrase in self.phrases:
            total += len(phrase)
        return total

    def __gt__(self, obj):
        return self.total_characters() > obj.total_characters()

    def __lt__(self, obj):
        return self.total_characters() < obj.total_characters()

    def __eq__(self, obj):
        return self.total_characters() == obj.total_characters()
```

Check It!

LAST RUN on 12/28/2023, 2:41:42 AM

Test passed

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